

Issues impeding the commercialization of laboratory innovations for energy-dense Si-containing lithium-ion batteries

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Silicon is a promising alternative to the conventional graphite anode in high-energy lithium-ion batteries owing to its high gravimetric capacity. However, intrinsic issues, such as severe volume expansion during cycling, have plagued the development of batteries that use Si anodes. While tremendous progress has been made in laboratories to tackle these issues, most Si-containing batteries in industry, in which Si anodes are made of Si suboxides or Si–C composites, can use only a very limited amount of Si. Here we review important factors that affect the practical energy density of Si-containing batteries, including electrode swelling and cut-off voltage in cell operation. We also discuss calendar life, safety and cost issues, which also have a strong influence on practical cell design. Furthermore, we propose testing protocols to evaluate the practical viability of newly developed Si anodes.

Conventional lithium-ion batteries have reached the limits of their energy density and cannot satisfy the rapidly increasing demands for long-range electric vehicles^{1,2}. Numerous attempts have been made to introduce silicon-based materials into conventional graphite anodes to increase the energy density because of the high gravimetric capacity of Si (3,592 mAh g^{−1})^{3–8}. However, blending Si-based materials (SiO_x and Si–C composites) with graphite has had only limited success in industry. The use of Si in commercial anodes is still highly limited (<5 wt%) due to performance degradation deriving from the intrinsic problems of Si anodes (Fig. 1)^{9–13}.

Most studies on Si-based anodes have focused on improving cycling performance under relatively mild testing conditions¹⁴. Practical testing conditions would have to include active materials with high mass loadings, low mass and volume fraction of inactive components such as binders and carbon additives, densely packed electrodes with low porosity, and lean electrolytes, to name but a few^{11,15}. Furthermore, calendar aging, storage conditions, stack pressure, safety and cost-effectiveness are also important factors for practical applications, but have rarely been considered in previous studies.

Despite increasing active collaborations between academia and industry to develop Si-based anodes, there is still no strategy for evaluating Si-containing cells with a comprehensive set of practical performance metrics. In addition, previous reviews have mainly concentrated on the progress in materials development^{16–18}, with few reports discussing in any detail the challenges of Si-containing cells from a practical viewpoint^{15,17}.

In academia, the energy density is often calculated on the basis of only the active materials, whereas in industry the energy density must be considered at the cell level^{19–23}. The values obtained without considering practical conditions, such as electrode expansion and inactive components, are artificially overestimated. In addition to this overestimation, there are several important factors that affect the practical application of Si anodes, such as the decrease in discharge cut-off voltage and calendar life, that could be easily overlooked by many researchers.

Here we consider a number of aspects of anodes that impede the development of practical Si-based batteries. First, we discuss the importance of electrode swelling in practical cell design, especially its

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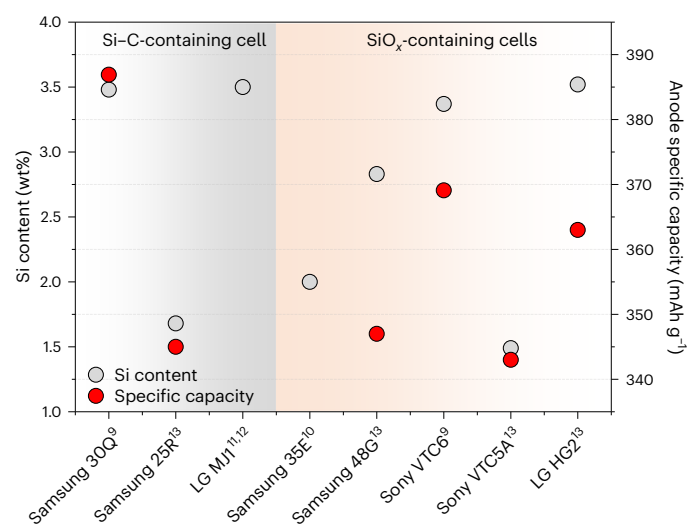


Fig. 1 | Specification of commercial Si-containing 18650 cells. The Si content and specific capacity of the anodes used in the commercial 18650 Si-containing cells of various battery companies^{9–13}. Si-C and SiO_x-graphite composites are the two main types of Si-based anode used in industry.

effect on the cell energy density. We then highlight the dependence of the cut-off voltage on the Si content in Si-based anodes. The calendar life, safety and cost-effectiveness of Si-containing cells are also considered, along with comparisons with graphite-based cells. Considering all these factors, we suggest general testing protocols to estimate the commercial feasibility of Si-based anodes.

Influence of electrode swelling on cell energy density

The cell energy density can be maximized when the active components occupy as limited a space of the cell as possible. However, in commercial cell design, any free space inside the hard or soft casing is often reserved for accommodating the swelling of the cell (Fig. 2a,b)^{9,24,25}. As the conventional graphite anode has a low electrode swelling of less than 20%, graphite-based cells require a relatively small free space to accommodate its swelling during battery operation²⁶. With its low swelling and small free space, the graphite-based cell can achieve a reasonable energy density despite its low theoretical specific capacity. However, Si-containing cells inevitably require a larger free space than graphite cells because of their larger electrode expansion. This would lower the energy density and attenuate the benefit of the high specific capacity of Si-based anodes. Most studies have overlooked the dependence of energy density on the free space inside the cell while recognizing electrode expansion.

In addition to the free space, the electrode density or electrode porosity should also be considered. Prado et al. showed that the swelling of a Si-graphite (30 wt% Si content) electrode (Fig. 2c) is highly dependent on its density²⁷. In general, a Si-containing electrode with a lower packing density (higher electrode porosity) shows a lower swelling because it has more room to accommodate the volume expansion^{28,29}. However, a higher electrode porosity requires not only a thicker electrode (to maintain the same loading) but also more electrolyte to fill the pores, which lowers the cell energy. It is noted that a commercial graphite or Si-containing anode generally has an electrode density of around 1.6 g cm⁻³ and an electrode porosity of around 30%. However, many studies on Si-based anodes have been performed with a higher electrode porosity to alleviate volume variation, which could severely lower the energy density.

To clarify the influence of electrode swelling on the energy density, we estimated the energy densities of commercial batteries consisting

of SiO_x or Si nanoparticles and graphite with varying content of Si. As an illustration, we chose a common Ni-Mn-Co cathode (NMC811) to construct pouch-type full cells. The pouch cells generally had dimensions of 102 mm × 354 mm × 9.5 mm and a capacity of over 50 Ah. The details of the cell parameters are provided in Source data Fig. 2 refs. 13,27,30–36. The calculation of energy density is described in Methods.

First, we confirmed the anode electrode swelling as a function of Si content, based on information on electrode swelling and porosity from the literature (Fig. 2d,g)^{27,36,37}. Si-blended electrodes have higher specific capacities than SiO_x-blended electrodes, but they also have higher porosities. Therefore, the two types of electrode have similar swelling.

We then plotted the volumetric energy density as a function of Si and SiO_x content (Fig. 2e,h). Cells with over 10 wt% SiO_x show a higher energy density than the commercial graphite cell. However, it is noteworthy that Si-containing cells with over 20 wt% SiO_x in the anode show reduced energy densities. This is because higher electrode swelling requires a larger free space, which reduces the number of stacks in the confined space of the pouch cell, thereby leading to reduced cell capacity. The cell with 60 wt% SiO_x in the anode has an energy density close to that of graphite, even though its specific capacity is more than three times higher than that of the graphite cell. Cells with over 70 wt% SiO_x show lower energy densities than the graphite cell. A similar trend is observed for the Si-containing cells; however, they all have a lower energy density than the graphite cell because of their high electrode porosity and swelling.

We also calculated the gravimetric energy density (Wh kg⁻¹) for each cell to observe the impact of Si incorporation into anodes (Fig. 2f,i). SiO_x-containing cells show higher gravimetric energy density than the graphite cell, with the gravimetric energy density increasing with Si content. Si-containing cells also exhibit an overall increasing trend in gravimetric energy, although their energy densities are lower than that of the graphite cell. Nonetheless, it should be noted that the magnitude of these increases in gravimetric energy density is quite small for both SiO_x- and Si-containing cells. As mentioned above, the anode loading must decrease substantially with increasing Si or SiO_x content to maintain a fixed capacity ratio of the anode and cathode (N/P). The energy density could be enhanced by increasing the number of electrode stacks. However, anode expansion prevents the addition of many stacks, thereby limiting the improvement in gravimetric energy density that can be achieved. For example, despite the large difference in the specific capacity of anodes containing 50 wt% and 100 wt% Si (or SiO_x), their severe expansion means that the maximum number of electrode stacks is the same. In addition, due to the low initial Coulombic efficiency of Si anodes, the areal capacity of the cell decreases as the Si content increases, which also hinders the improvement in gravimetric energy density.

Although these results might be different from those obtained from cells used in practical applications, they indicate the importance of electrode swelling and porosity for cell-level energy density. In addition, these estimations imply that a higher specific capacity of an anode does not guarantee a higher cell energy. In this regard, we encourage researchers to carefully evaluate the performances of Si-based anodes by considering practical cell conditions.

The observation of anode thickness variation during repeated cycling could be a good indicator of a stabilized electrode swelling. After an initial swelling of 15%, graphite presents stable thickness variation without further increases in overall thickness during extended cycling. However, unlike graphite, Si-containing cells show gradually increasing electrode swelling during repeated cycling, even after a large electrode expansion in the initial cycle, because of irreversible expansion of the Si-containing electrode after the fully discharged state. This irreversible swelling results from a morphological change in Si, accumulation of the solid-electrolyte interphase (SEI) layer,

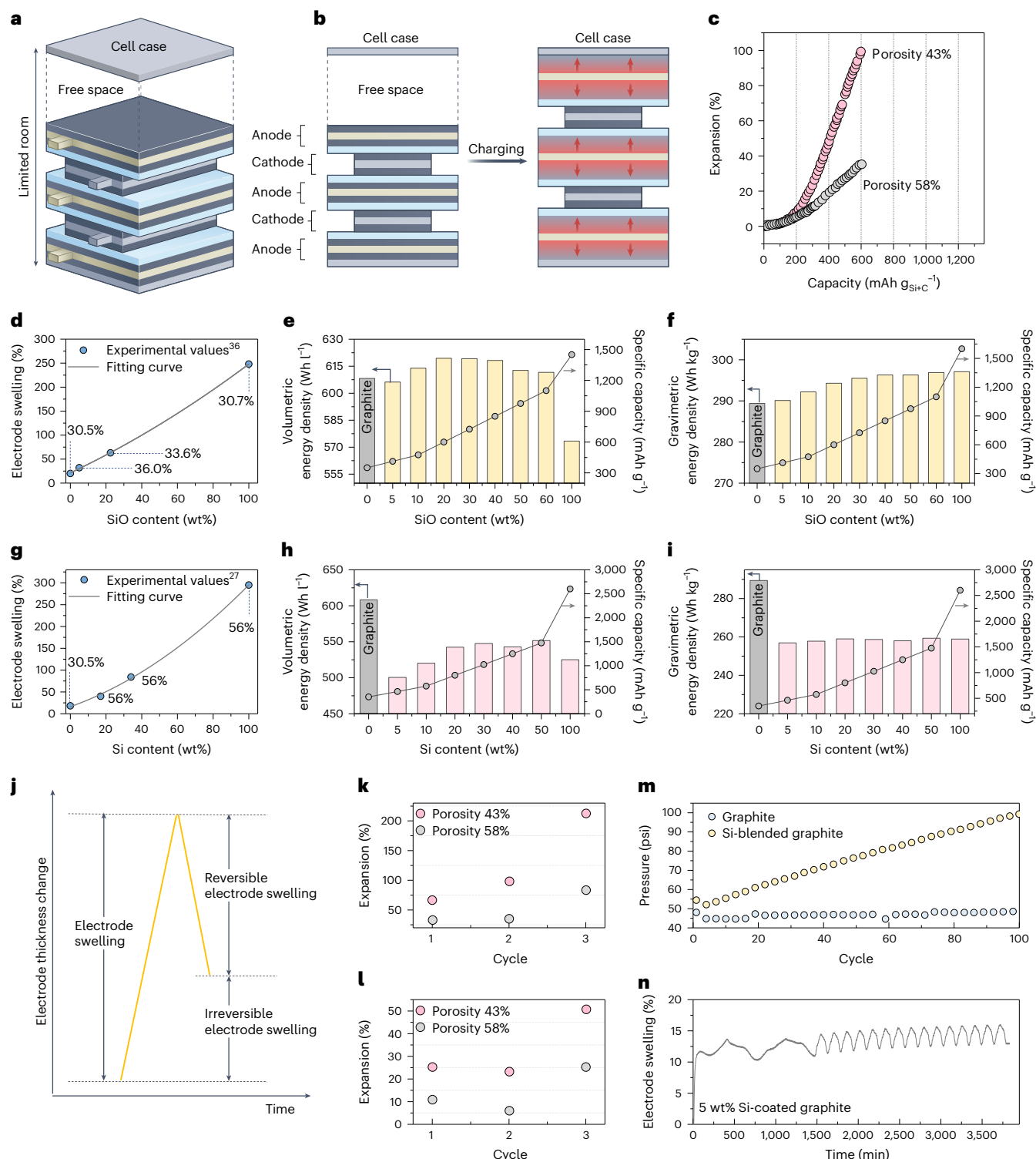


Fig. 2 | Volume expansion of Si anodes and related issues. **a**, Schematic illustration of a stacked cell with the space margin denoted as free space. The blue layer between the cathode and anode represents the separator. **b**, Cross-section of the stacked cell. Upon charging, the free space is used to accommodate the swelling electrode, indicated by red arrows. **c**, Electrode swelling as a function of the lithiation capacity of the Si-containing graphite electrode (30 wt% Si nanoparticles). Two different electrode porosities were considered²⁷. **d–i**, Electrode swelling (**d,g**) and estimated cell-level volumetric (**e,h**) and gravimetric (**f,i**) energy densities along with the specific capacity as a function of SiO (**d–f**) or Si content (**g–i**) in the anode. Electrode porosity values (%) are given in **d** and **g**. The data were fitted by the polynomial curve fitting method. The data on electrode swelling, porosity and specific capacity are

taken from the literature^{27,36}. **j**, Schematic plot showing the change in electrode thickness due to both reversible and irreversible electrode swelling during cycling. **k,l**, Maximum (**k**) and irreversible (**l**) electrode swelling of a Si-containing anode (30 wt% Si nanoparticles) with different electrode porosities in the first three cycles²⁷. **m**, Average cell pressure of a graphite cell and a Si-blended graphite cell during cycling³⁹. **n**, Real-time electrode swelling behaviour of a Si-graphite anode in a full cell during cycling⁴⁰. The volumetric and gravimetric energies were calculated on the basis of our model cells; the cell specification and parameters are summarized in the Source data and details of the calculations are summarized in Methods^{13,27,30–36}. Panels adapted with permission from: **c,k,l**, ref. 27, under a Creative Commons licence CC BY 4.0; **m**, ref. 39, Elsevier; **n**, ref. 40, Springer Nature Ltd.

particle detachment in the electrode and delamination from a current collector (Fig. 2j). On account of this, a Si-based anode might show gradually increased electrode swelling over extended cycling even if it exhibits only a small expansion in the initial cycle. Prado et al. demonstrated the irreversible expansion behaviour of a Si-graphite electrode with 30 wt% Si (Fig. 2k,l)²⁷: the Si-graphite electrode showed increased expansion with cycling, and the irreversible swelling also increased with cycling, leading to a permanent thickness expansion after full delithiation.

In practical cell applications, a certain amount of stack pressure is typically applied to the module. The external pressure can suppress irreversible expansion resulting from interparticle detachment and electrode delamination and thereby alleviate the impact of electrode swelling on the energy density³⁸. However, the external pressure can bring about extremely high compressive stress in the Si-containing cell due to the continuously increasing swelling during cycling. The external pressure and high stress in the cell result in lowered electrode porosity and increased ionic pore resistance during cycling, leading to capacity fading. Louli et al. conducted operando pressure measurements to determine the effect of irreversible swelling on the performance of various Si-containing cells³⁹. Their study revealed that while graphite cells maintained almost constant pressure at discharge, Si-containing cells showed continuously increasing cell pressure during cycling (Fig. 2m), which can cause fatal cell damage. For this reason, it is important to monitor electrode thickness to ascertain whether the Si-based anode reaches a stabilized swelling point within a few cycles (Fig. 2n)⁴⁰. If the electrode swelling cannot be stabilized, it should not be used as the anode in real batteries. To ensure cell stability, the electrode swelling at the stabilized point, and not in the first cycle, should be considered in the design of Si-containing cells. Otherwise, cells may bulge or burst by excessive electrode expansion during cycling.

In addition to irreversible expansion, anodes can suffer in-plane (*x* direction) swelling as well as through-plane (*z* direction) expansion. Pietsch et al. reported that a graphite anode shows a small amount of in-plane expansion during charging as the particle shape and expansion orientation allow the electrode to expand in all directions⁴¹. The in-plane swelling of graphite is negligible compared with that of through-plane swelling. However, in the case of Si-based anodes, the in-plane swelling can be severe due to the large anisotropic expansion of Si (ref. 42). The in-plane expansion can bring about fatal issues, including electrode delamination and short circuiting by contact between the anode and cathode. Despite its importance, this in-plane expansion has been overlooked in academia owing to a lack of awareness of this problem.

Cut-off voltage and its dependence on Si content

Figure 3a shows the delithiation voltage profiles of anodes prepared with graphite, Si-coated graphite (5 wt% Si) and Si-coated carbon (35 wt% Si) from the literature⁴⁰. The voltage profile is more sloping for the Si-containing anodes compared with the graphite anode due to the relatively high working potential and amorphous nature of Si. This slope varies depending on the properties of the Si material^{37,40,43,44}. As shown in Fig. 3a, most of the overall capacity of graphite is provided in the low voltage region (below 0.3 V), while Si-containing anodes have a substantial capacity contribution at until 1.5 V. The sloping voltage profiles of the Si-based anodes means that a higher delithiation voltage cut-off is needed to use their maximum capacities compared with graphite. This also means that in the full-cell configuration, Si-containing cells require a lower cut-off voltage on discharging than the graphite cell to realize their full capacity.

Figure 3b shows the anode voltage cut-offs as well as the corresponding full-cell voltage cut-offs from previous studies of Si-containing cells^{9,45–48}. Each cut-off voltage for the Si-based anodes was measured by full-cell testing using a three-electrode configuration. The anode including around 15 wt% Si has an end-of-discharge

voltage of around 0.68 V for a 3.0 V full-cell cut-off. In the case of the 2.8 V full-cell cut-off, the anode voltage ends at 0.83 V. These voltages (0.68 and 0.83 V) are too small to use the full capacity of the Si-based anode. For the 2.5 V cut-off, the end-of-discharge voltage of the anode is around 1.12 V with >15 wt% Si content and 1.09 V with <5 wt% Si content. Regardless of the Si content, the end-of-discharge voltage is similar in all cases. As shown in Fig. 3a, because a higher Si content corresponds to a more sloping voltage plateau, the cell with a lower Si content can provide a capacity closer to its maximum value than the cell with a higher Si content. In the case of a Si-dominant cell (>85 wt% Si content), its end-of-discharge voltage is around 0.91 V under a 2.3 V cut-off. This relatively low discharge voltage is not sufficient to fully use the capacity of the Si-dominant anode (Fig. 3a). These results indicate that the Si-containing cell requires a much lower full-cell cut-off voltage than the graphite cell to use its maximum capacity. Additionally, it should be noted that these cut-off voltage behaviours are dependent on the cell design, including the N/P ratio, cell dimensions and targeted energy density. The inventory of active Li ions also varies depending on the cell design, even when using electrodes composed of the same materials, leading to different cut-off voltages.

We calculated the full-cell discharge voltage profiles of cells containing different Si contents by subtracting the voltage profiles of various anodes, reported in previous studies^{30,40}, from that of a $\text{LiNi}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}\text{O}_2$ (NMC622) cathode (Fig. 3c). As the Si content in the anode is increased, the slope of the voltage profile becomes more pronounced. To verify the effect of added Si on voltage behaviour, we confirmed the required discharge cut-off voltage to achieve 99.0% of the theoretical cell capacity (Fig. 3d). The discharge cut-off voltage was lowered from 3.13 to 2.49 V on incorporating more Si into the electrode. In addition, we calculated the average voltage for each cell (Fig. 3d) using the voltage profiles shown in Fig. 3c. The average voltage of the cells gradually decreased from 3.68 to 3.39 V with increasing Si content in the anode, with the voltage of the cell with 35 wt% Si in the anode being 8% lower than that of the graphite cell. The reduced voltage not only lowers the cell energy, but also attenuates the benefit of using high loadings of Si in the anode. Furthermore, the low discharge cut-off voltage increases the anodic potential and consequently induces side reactions such as Cu dissolution from the anode current collector^{49,50}. Cu dissolution could lead to severe problems, including additional SEI formation, increased cell resistance, non-uniform current density and safety issues. Hence, the cut-off voltage of Si-containing cells is critical because it can directly affect cell performance. Nonetheless, this issue has not been considered in detail in most of the studies related to the development of Si materials.

Importance of calendar life and its evaluation

Calendar aging, a time-dependent performance fade, is an intrinsic behaviour that must be considered to estimate the long-term health of the Si-containing cell. However, most research has focused on strategies to improve the charge–discharge cycle life of Si, with only a few studies on the calendar life of Si-containing cells⁵¹. The conventional graphite cell can achieve over 15 years (time to reach 80% of cell capacity) of calendar life under storage conditions of 20–40 °C, while a Si-containing cell typically exhibits a calendar life of less than 1 to 2 years, even with a low Si content⁵². According to the United States Advanced Battery Consortium (USABC), a calendar life of over 5 years is advisable for electric vehicle batteries, and the criterion should be extended for batteries requiring longer lifetime such as large-scale energy storage systems⁵³. While the electrochemical cycle properties and energy densities of state-of-the-art Si-containing cells are rapidly approaching the target values set by the United States Department of Energy (DOE)⁵⁴, there is still a huge gap between the calendar life of Si-containing cells and the DOE's target value. This gap implies that the strategies to enhance cycle life may not substantially improve time-dependent fade. The short calendar life is one of the major roadblocks to the practical application of Si-containing cells.

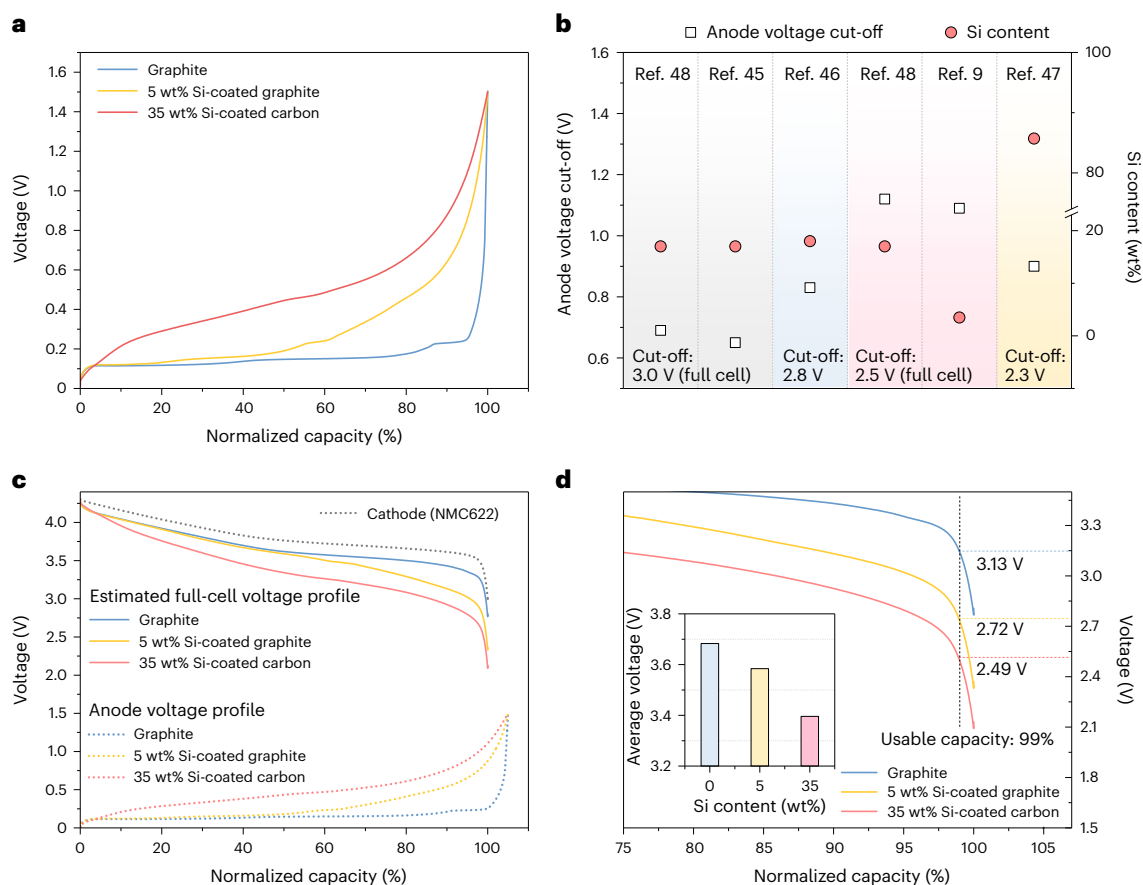


Fig. 3 | Lower cut-off voltages as Si is added to graphite anodes. a, Normalized delithiation voltage profiles for graphite, Si-coated graphite (5 wt% Si) and Si-coated carbon (35 wt% Si) anodes with a specific capacity of about 360, 510 and 1,250 mAh g⁻¹, respectively⁴⁰. **b**, Anode voltage cut-off and the corresponding Si content in the anode in various full cells with different full-cell cut-off voltages^{9,45–48}. **c**, Voltage profiles of full cells with different Si contents, estimated

by subtracting the voltage profiles of various anodes from that of the NMC622 cathode^{30,40}. **d**, Magnification of the voltage profiles in **c**, showing the discharge cut-off voltage required to obtain 99% of the usable cell capacity. Inset: the average discharge voltage of each full cell. Panels adapted with permission from: **a,c,d**, ref. 40, Springer Nature Ltd; **c**, NMC622, ref. 30, Wiley.

In general, cycle life aging of Si-containing cells mainly originates from undesired chemical reactions between a newly exposed anode surface (caused by SEI fracture) and electrolytes, continuous electrolyte decomposition and mechanical degradation. Calendar life aging correlates with the stability of the surface passivation (SEI) layer and the Si underneath it towards electrolytes during long-term storage^{55,56}. Unfortunately, the passivation layer on the Si surface is unstable owing to the higher chemical reactivity of lithiated Si compared with graphite at equivalent potentials^{57–59}. The charged Si tends to reduce and decompose the binders, Li salts and organic solvents in the electrolyte, leading to the reconstruction of SEI layers with a loss of cyclable lithium content^{60,61}. The loss of lithium inventory from Si leads to self-discharge and contraction of the lithiated Si particles, which results in mechanical disruption of the SEI layers. An additional SEI layer can also be formed on the newly exposed Si surface caused by the self-discharge, which leads to the further loss of cyclable lithium. Furthermore, Si-containing cells are highly vulnerable to the hydrofluoric acid (HF) generated by the hydrolysis of common Li salts such as LiPF₆. A conventional graphite cell typically shows moderate tolerance towards HF without considerable cell degradation^{62,63}. However, Si materials can be etched by HF, leading to the loss of active Si material and the generation of H₂O, gases and soluble Si species (SiF₄ and H₂SiF₆)⁶⁴. These by-products accelerate HF production, block the pore network of both the anode and cathode, and increase impedance by forming a passivation layer on the cathode surface. The mechanisms of calendar aging are depicted

in Fig. 4a. These degradation phenomena can be accelerated when a cell is in a higher state of charge or stored at a higher temperature^{65,66}. In this context, pre-lithiation treatment of a Si-containing anode may improve its cycle life, but at the same time may lead to worse calendar aging because of the high reactivity of lithiated Si.

To evaluate calendar aging, long-term experiments are required that could involve a testing period of over a year, such as the testing procedure established by the USABC⁵³. This long-term testing is practically difficult in a laboratory environment. As an alternative, methodologies based on potentiostatic hold have been developed to measure the loss of lithium inventory to SEI layers in Si-containing cells in experiments of relatively short duration. Figure 4b shows the voltage hold testing profile established by the DOE Silicon Consortium Project⁶⁷. It should be noted, however, that voltage hold cannot replace the traditional long-term test for calendar aging. Schulze et al. have discussed the principles and methodology of the potentiostatic hold test as an accelerated predictor of calendar life for various anodes, including graphite and Si (ref. 68). They measured the current response comprising contributions from both reversible and irreversible capacity, which implies continuous charging of the cell and irreversible loss of lithium (that is, parasitic reactions such as SEI formation). Kalaga et al. evaluated the calendar life of cells with anodes containing 15 wt% Si and compared it with a graphite cell⁶⁹. This study clearly showed that the addition of Si markedly increases the integrated parasitic current compared with the Si-free graphite cell (Fig. 4c). Moreover, the addition

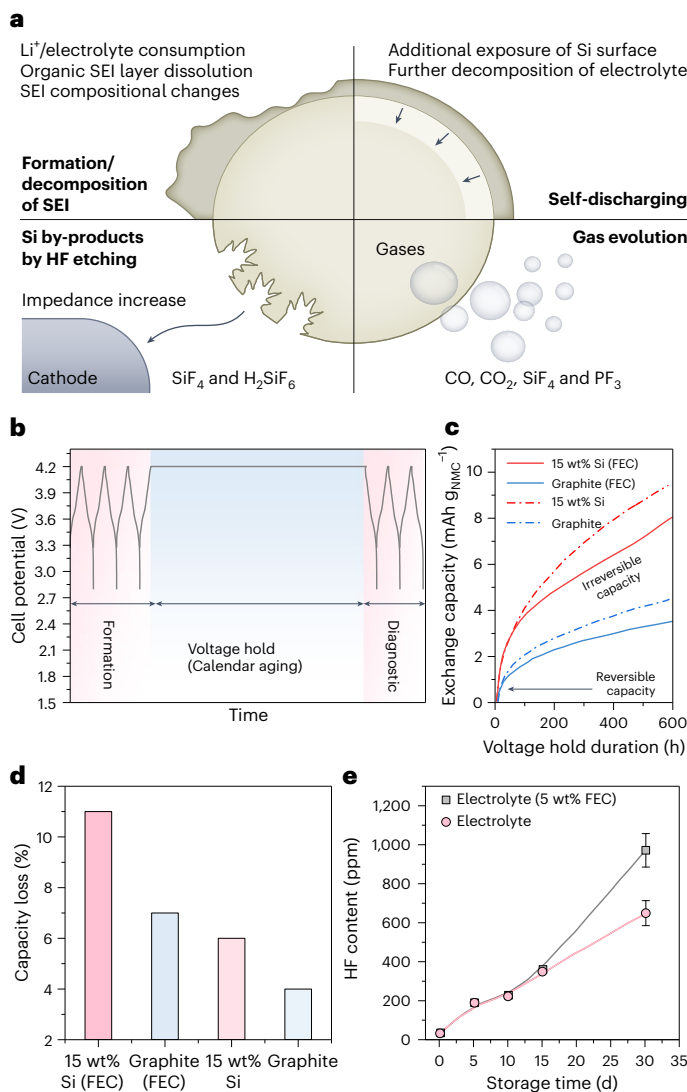


Fig. 4 | Mechanism and analysis of calendar life aging of the Si anode.

a, Schematic illustration of various degradation phenomena in a Si-containing anode on calendar aging. The damaged beige sphere represents an aged Si particle and the grey layer is the SEI. **b**, Voltage hold testing protocol established by the US DOE Silicon Consortium Project⁶⁷. The exchange capacity can be obtained from the parasitic reaction (irreversible capacity) and residual charging (reversible capacity) depending on the voltage holding time. **c**, Integrated parasitic capacity during calendar aging of a pure graphite cell and a Si-containing graphite cell (15 wt% Si) with and without FEC additive in the electrolyte⁶⁹. **d**, Capacity loss of full cells following three diagnostic cycles after voltage hold for 600 h (ref. 69). **e**, HF generation in electrolytes with and without FEC during thermal aging for 30 days at 60 °C (ref. 62). Panels adapted with permission from: **b**, ref. 68, under a Creative Commons licence CC BY 4.0; **c**, **d**, ref. 69, Elsevier; **e**, ref. 62, IOP.

of fluoroethylene carbonate (FEC) seems to suppress the side reaction in both cells during voltage hold. However, the capacity loss after calendar aging of the Si-containing cell with FEC was almost twice that of the Si cell without FEC (Fig. 4d). These opposite results for the integrated parasitic current during voltage hold and capacity loss after calendar aging can be explained by the oscillating particle size of Si (ref. 69). During the voltage-holding step, the Si particles remain intact without any substantial change and the SEI layers are relatively stable towards mechanical disruption such as from contracting Si particles, which can effectively block HF attack on Si. However, as the cell is discharged after the voltage-holding step, the SEI layers collapse due to the contraction

of the Si particles. The accumulated HF then penetrates the SEI layers and reacts with Si, leading to severe degradation. It is well known that FEC can produce HF by dehydrofluorination at elevated temperatures. Shin et al. confirmed that a cell including 5 wt% FEC in the electrolyte produces more HF than that without FEC after 30 days of thermal aging at 60 °C (Fig. 4e)⁶². This indicates that the FEC additive induces a much higher capacity loss in Si-containing cells after calendar aging at elevated temperatures. Considering that FEC is commonly used as an additive to improve the cycle life of Si cells, stability issues affecting the calendar life of cells including FEC present another critical challenge for the design of Si-containing cells.

Safety concerns

Adopting Si-based materials in anodes may create safety concerns as the large volume change and chemical instability of Si materials during the (de)lithiation process could induce cell burst and thermal runaway. As we described briefly above, the gassing behaviour of Si-based anodes can be more severe than that of conventional graphite anodes because of the unstable passivation layer derived from the high chemical reactivity of lithiated Si. Typically, the gas generated from initial cycling is vented by a degassing process. However, due to undesirable decomposition reactions caused by the intrinsic reactivity of Si, gases can be continuously generated during cell operation. Seitzinger et al. explored the chemical reactivity and gassing behaviour of Si anodes as a function of lithium salt and surface chemistry of Si (Fig. 5a)⁵⁵. They found that all Si anodes terminated with Si (PECVD, Nanoamor and S5505) generated various gases originating from the dissolution of Si and decomposition of the electrolyte and lithium salt. Aggressive gas evolution can cause the cell to bulge or burst when the cell has a confined casing, resulting in the exposure of reactive electrodes and flammable carbonate electrolytes to air (Fig. 5b). These gassing behaviours may contribute to the thermal runaway of cells. In addition to gas evolution, electrode swelling during cycling can cause distortion or bulging of the electrode, resulting in fatal safety issues such as internal short circuiting (Fig. 5c)⁹.

Although Si materials normally exhibit good thermal stability comparable to that of graphite, an excessive amount of SEI on the surface of Si anodes during repeated cycling can lead to much higher heat generation because of the exothermal nature of SEI decomposition⁷⁰. Park and Lee analysed the thermal stability of lithiated Si as a function of Si particle size using differential scanning calorimetry (DSC; Fig. 5d,e)⁷¹. The analysis showed that electrodes prepared with smaller Si particles tend to generate a larger amount of heat at an earlier stage (when heated to 300 °C) and have a lower onset temperature of heat generation due to the accumulation of SEI layers on the larger surface area of Si (Fig. 5e). As is well known, reducing the particle size of Si is a common approach to addressing the intrinsic issues caused by the volume variation of Si and improving its cycling stability. However, it should be noted that this approach can also raise safety concerns owing to the greater surface area (that is, the large amount of SEI) of the smaller-sized Si particles, because abnormal heat generation can trigger the decomposition of the pre-existing SEI, which in turn can produce more heat and ultimately lead to the melting of the polymeric separator and then an internal short circuit. This highlights that the trade-off between improved performance and safety issues requires careful consideration when developing Si-based anodes. The chain of events originating from the intrinsic problems of Si-based anodes and resulting in various safety concerns are summarized in Fig. 5f.

Cost-ineffectiveness of cell Si addition

As shown in Fig. 2, adding a rational amount of Si to graphite anodes can enhance the cell energy density. However, the increased energy density does not guarantee the cost-effectiveness of Si-containing cells in terms of the US\$ per kilowatt hour metric. Undoubtedly, performance factors such as energy density, cycle life and calendar life are highly important

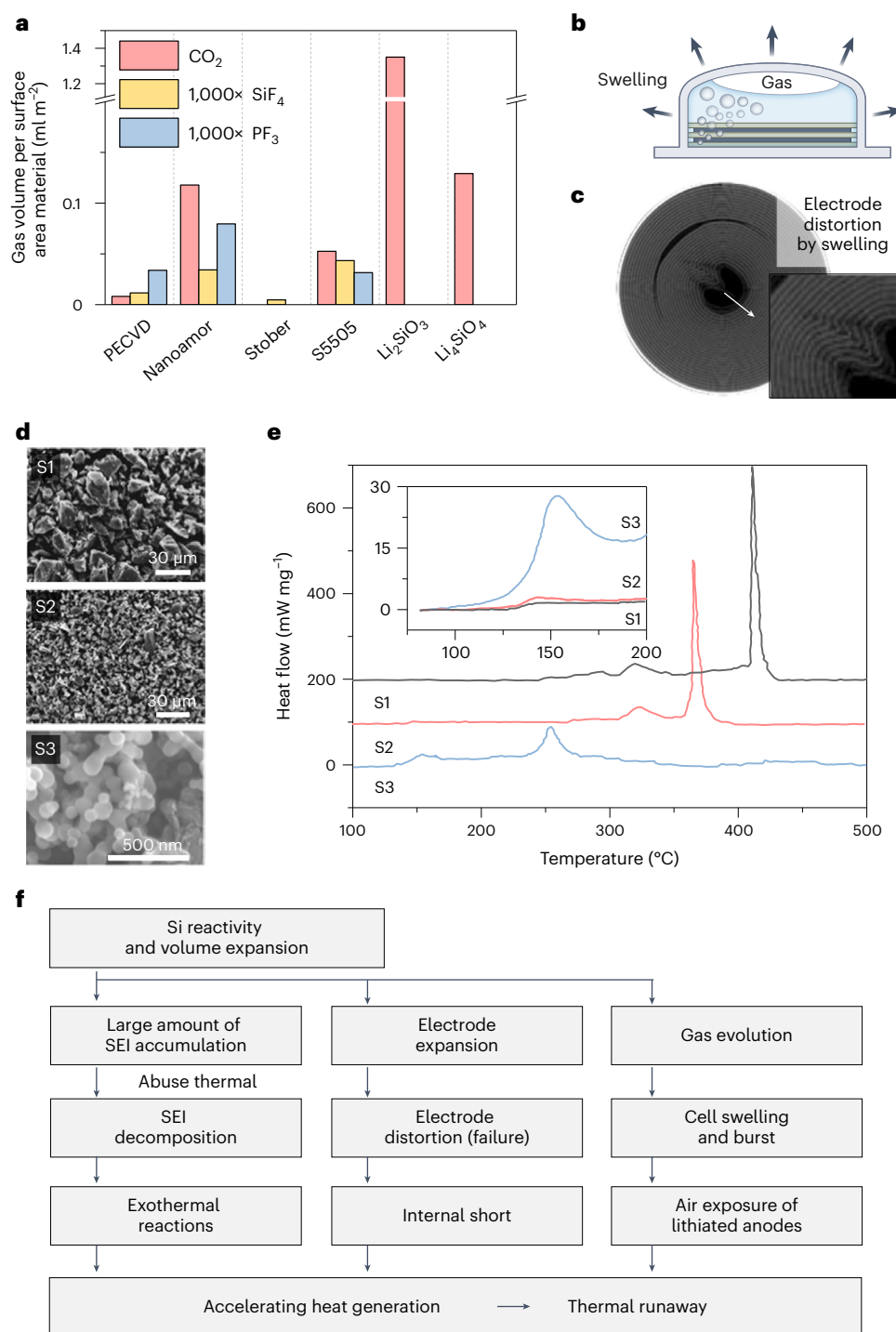


Fig. 5 | Safety issues derived from Si-based anodes. a, Gas volume generated per surface area of Si-containing materials during 14 h. **b**, Schematic illustration of a bulging pouch cell as a result of gas evolution from the reactions between Si and electrolytes⁵⁵. Light grey, pouch casing; blue, electrolyte; green, anode; dark grey, cathode. **c**, Computed tomography measurement of a cylindrical cell containing Si-based anodes during cycling⁹. Inset: electrode distortion caused by swelling. **d**, SEM images of Si powders with a particle size of 15, 5 and

0.03–0.050 μm , denoted as S1, S2 and S3, respectively⁷¹. **e**, Corresponding DSC curves for the fully lithiated state of electrodes prepared with Si powders S1, S2 or S3 (ref. 71). The inset shows the curves on a larger scale. **f**, A flow chart illustrating the thermal runaway mechanism in Si cells. Panels adapted with permission from: **a**, ref. 55, American Chemical Society; **d,e**, ref. 71, Korean Chemical Society. Panel **c** reproduced with permission from ref. 9, Elsevier.

for the practical application of Si cells, but cost-effectiveness is also an essential factor to be considered for the commercial viability of Si cells. As is well known, the cost of a conventional graphite anode is a relatively small proportion (<15%) of the total cell cost due to its moderate price, while a typical Ni-based layered oxide cathode accounts for over 50%

of the total cell cost⁷². For this reason, the cost of graphite-based cells has mainly been focused on the cathode side. In the case of Si, the raw material cost of bulk Si (<5 US\$ kg^{-1}) is lower than that of commercial graphite (<15 US\$ kg^{-1})^{73,74}. However, the Si materials used in anodes demand complicated and costly synthetic methods to mitigate their

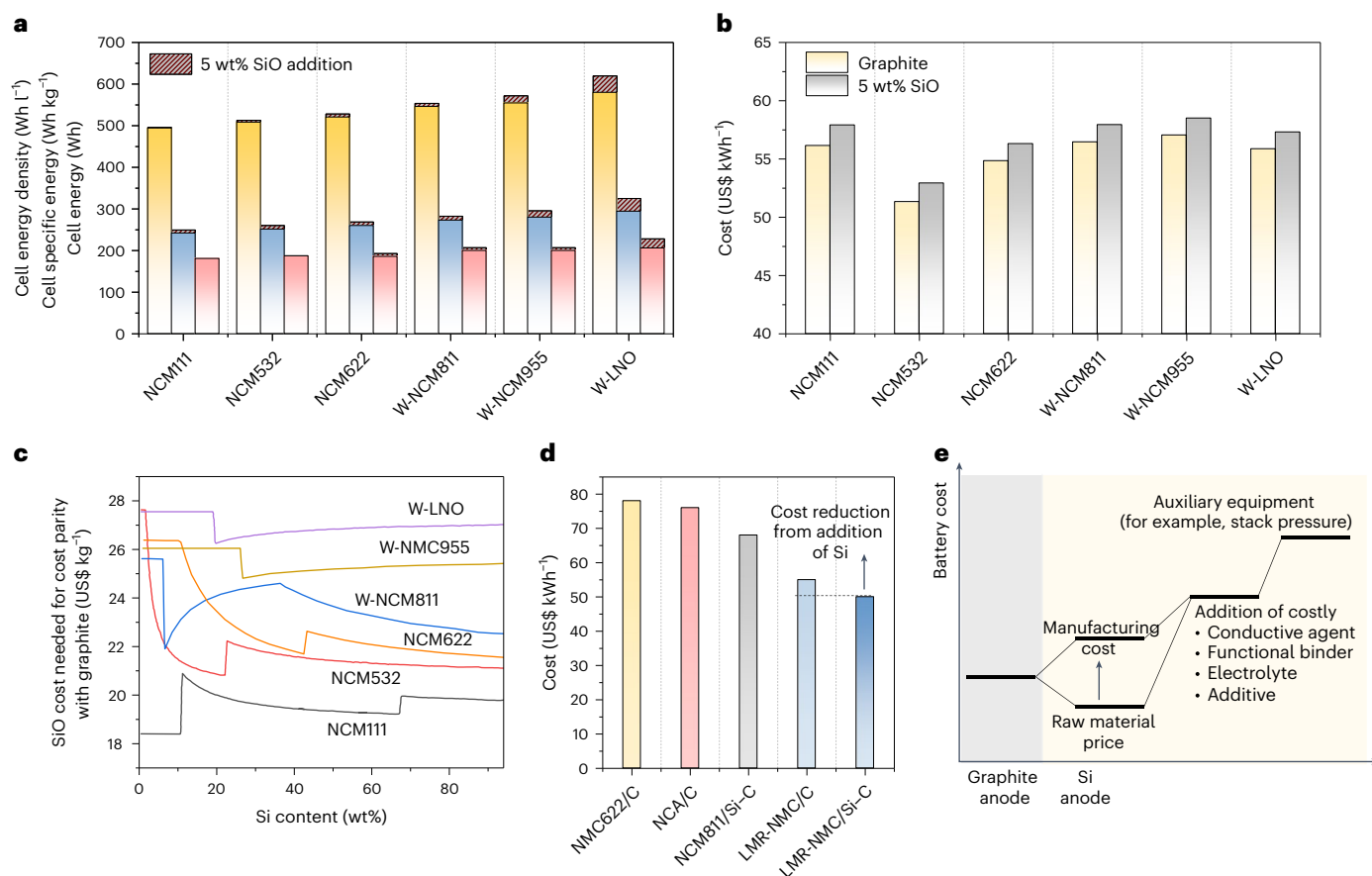


Fig. 6 | Cost of Si-containing cells. **a**, Energy density (yellow), specific energy (blue) and cell energy (pink) for various full cells. The anode with 5 wt% SiO is paired with NCM111, NCM523, NCM622, W-NCM811, W-NCM955 and W-LNO cathodes. Tungsten-doped cathodes are indicated by the prefix W (ref. 75). **b**, Effect of SiO addition into the anode on cell-level cost. All cell components were considered in the cost calculations⁷⁵. **c**, Maximum cost of SiO anodes to attain cost parity with a cell using only graphite as a function of SiO content⁷⁵.

d, Cost estimations of cells with different anode and cathode compositions at the electrode stack level¹. NCA, $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$; LMR, lithium or manganese-rich layered cathode. **e**, Graph showing an estimation of battery cost based on a high manufacturing cost of Si, the addition of costly inactive materials (for example, conductive agent, binder, electrolyte and additive) and auxiliary equipment. Panels adapted with permission from: **a–c**, ref. 75, Elsevier; **d**, ref. 1, Springer Nature Ltd.

intrinsic problems mainly related to volume variation. These multi-stage and costly methods substantially increase the overall cost of cell production.

SiO_x is widely used in commercial Si anodes and can be synthesized through mature technologies. Greenwood et al. recently reported a detailed bottom-up performance-and-cost assessment of pouch-type Si-containing cells⁷⁵. By measuring the cell-level energy density of full cells containing 5 wt% SiO_x in graphite-based anodes and various cathodes, they showed that the introduction of this small amount of SiO_x leads to an increase in energy density of up to 7.62 Wh l^{-1} , even after taking into account electrode swelling (Fig. 6a). Increasing the content of SiO_x does not necessarily bring about energy gains depending on the cell design, which is consistent with the results shown in Fig. 2. In terms of cost-effectiveness, the addition of 5 wt% SiO_x was not more beneficial than the graphite cell when applying a SiO_x price of $60 \text{ US\$ kg}^{-1}$ (Fig. 6b). According to the modelling of Greenwood et al., each 1 wt% SiO_x addition increases the anode cost by approximately $0.53 \text{ US\$ kg}^{-1}$. This cost-ineffectiveness would be worse when using more SiO_x to achieve a higher energy. Figure 6c shows the target cost of SiO_x to achieve cost parity with graphite as a function of SiO_x addition⁷⁵. These costs were modelled with an optimistic cathode capacity, and the values may be different from actual batteries depending on the cell design. Meanwhile, Schmuch et al. estimated that the addition of Si–C composite (20 wt% Si) is more cost-effective than conventional graphite anode-based

cells (Fig. 6d)¹. This different trend in cost arises from the much lower material cost of Si–C composite ($25 \text{ US\$ kg}^{-1}$) compared with that used by Greenwood et al. in their study ($60 \text{ US\$ kg}^{-1}$). This low price of Si indicates that the addition of cheap Si with high specific capacity can be beneficial in reducing the cost of the cell. However, although the raw material cost of Si is normally cheap, the fabrication of Si anodes may require additional processes such as nano-engineering and the construction of composites with carbon or an inactive matrix to overcome the intrinsic problems of Si, such as large volume expansion and low electric conductivity, raising the overall cost. Alternative methods to address these issues typically require the use of inactive components such as functional binders, electrolytes, additives and costly conductive carbon^{76,77}, absent in graphite cells, also increasing the material cost (Fig. 6e). Still further costs could arise from auxiliary equipment such as devices for applying surface pressure to balance cell thickness variation in actual cell operation³⁸. For these reasons, although it is difficult to precisely calculate the cost of newly developed Si materials in laboratories, we encourage battery researchers to carefully consider the cost of the raw materials used, the technological maturity of synthetic methods and the mass producibility of their materials.

Testing the practical viability of Si anodes

To encourage battery researchers to consider the factors discussed above, we propose testing protocols for estimating the practical

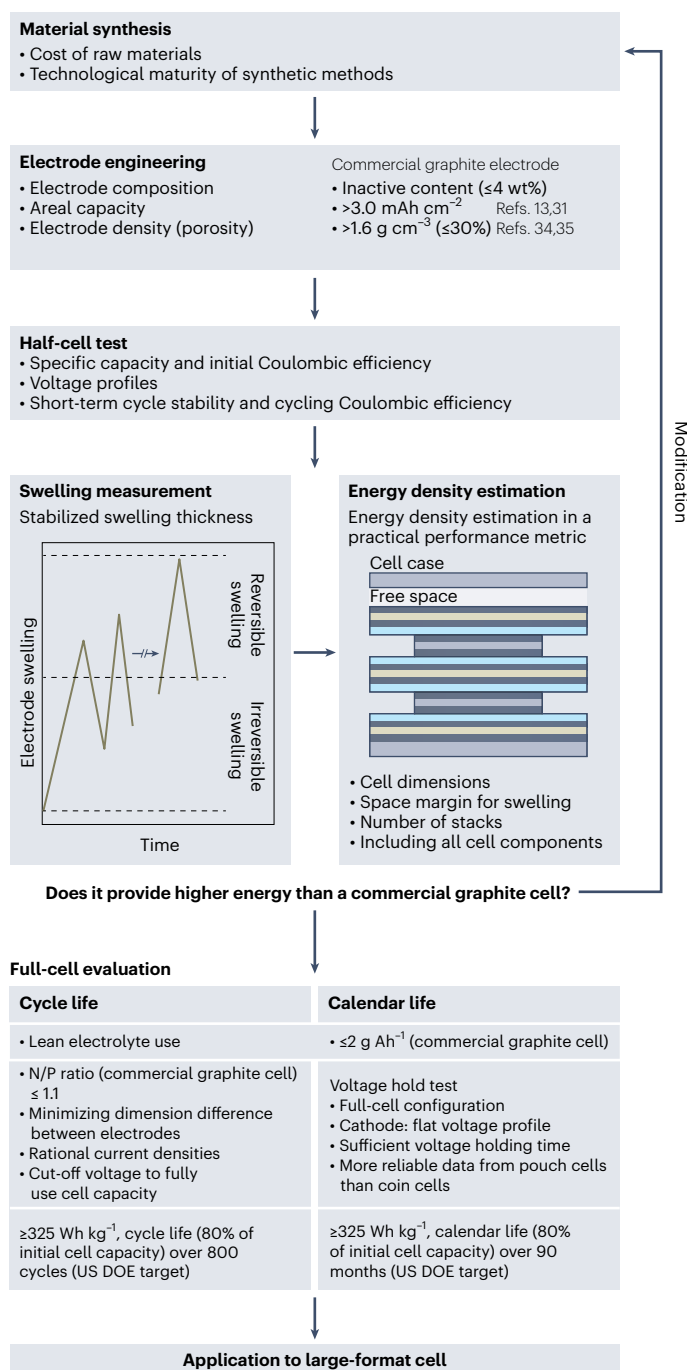


Fig. 7 | Testing protocol for Si-based anodes. A flow chart showing a testing protocol for evaluating Si-based anodes. The grey box in the 'Electrode engineering' box shows the specifications of a commercial graphite cell^{13,31,34,35}. The specification needs to be considered when developing feasible Si-based anodes with high energy density. The plot and schematic in the 'Swelling measurement' and 'Energy density estimation' boxes indicate that the free space within the cell dimensions for energy density estimation should be considered by measuring the changes in electrode thickness arising from both reversible and irreversible swelling until the thickness variation is stabilized. The grey boxes in the 'Full-cell evaluation' box exhibit the amount of electrolyte ($\leq 2 \text{ g Ah}^{-1}$) and N/P ratio (≤ 1.1) adopted in commercial cell design for cell evaluation. Such industrial evaluation conditions should be considered when assessing practical Si-containing cells at the laboratory scale.

viability of a newly developed Si anode on a laboratory scale (Fig. 7). First, for the synthesis of Si anodes, we recommend that researchers carefully consider the cost of raw materials and the technological

maturity of the synthetic methods they use. Even if a Si anode shows outstanding electrochemical performance, its industrial application could be difficult if material or processing costs are high.

Second, electrode specifications, including composition, areal capacity and porosity, should be set as competitive as those of commercial graphite anodes. The electrode specifications of commercial graphite cells are highlighted in the grey boxes in Fig. 7 refs. 13,31,34,35. The use of a Si anode might require more inactive components, such as binders and conductive carbon, than graphite because of its volume variation and low electrical conductivity, which can lower the cell energy density. High areal capacity is also needed to achieve a decent energy density, which can be achieved by minimizing inactive cell components such as current collectors. As shown in Fig. 2, electrode porosity also affects the cell energy density. It is important to set appropriate values for electrode porosity to maximize the cell energy density. In half-cell tests, it is difficult to determine whether the cycle stability of a Si anode is good because the half-cell contains excess lithium and electrolyte. We therefore encourage researchers to use half-cell test results simply to obtain basic performance information on Si anodes, such as reversible capacity, initial Coulombic efficiency and short-term cycle life.

The third step is to measure electrode swelling over extended cycling to identify its stabilized value, which can then be used to calculate the space margin inside a cell casing. The free space is the sum of the electrode thickness increment derived from both reversible and irreversible swelling until the swelling is stabilized. We suggest measuring the swelling by in situ thickness measurement by cell dilatometry or by using a micrometer gauge after cell disassembly in the fully lithiated state rather than by cross-sectional scanning electron microscopy (SEM). Electrode thickness observed by SEM can be easily misgauged by sample damage or a localized observation area.

Fourth, to estimate the energy density at the cell level, it is necessary to provide information on electrode densities, electrode and separator porosity, and areal mass loading. As we discussed earlier, maximum electrode thickness after swelling should be considered when determining the number of electrode stacks because of the severe swelling of Si-containing cells. Note that each battery company can vary these cell design parameters. To demonstrate the practical viability of Si materials, we also encourage comparison of the energy density of Si cells with that of state-of-the-art commercial graphite cells with high mass loading, low inactive content and low electrode porosity.

Last, full-cell tests are required to estimate cycle performance and calendar life aging. We recommend using a limited amount of electrolyte (approximately below $\leq 2 \text{ g Ah}^{-1}$) for the full-cell test for reliable results. In a commercial cell design, a minimum amount of electrolyte is generally injected into the cell to fill only the pore volume of electrodes and separator. An excess of electrolyte can produce seemingly promising performances, but a large amount of electrolyte not only lowers the cell energy but also requires substantial volume in the cell where the space is limited. In addition to the amount of electrolyte, it is also important to use an appropriate N/P ratio (≤ 1.1 for commercial graphite cells), minimize dimension difference between electrodes, apply rational current densities for cycling and establish a sufficiently wide voltage cut-off range to fully use the cell capacity. A high N/P ratio and a large deviation between electrode dimensions would lower the energy density because of the decrease in the use of active material. As we discussed earlier, an appropriate cut-off voltage range needs to be identified for a given Si content to use the full capacity of Si (Fig. 3). We recommend checking whether an experimental capacity value is close to the expected capacity calculated from the full-cell design. The performance targets set by the US DOE require that the end of life (80% of capacity retention) be over 800 cycles with a specific cell energy beyond 325 Wh kg^{-1} (ref. 54). We would like to remind researchers of both the cycle life and energy density targets in their full-cell evaluations.

When it comes to calendar life aging, a year-long experiment might be required to obtain meaningful data⁷³. Such long-term testing is difficult to perform in lab-scale experiments. There are several reliable alternative methods for estimating calendar aging based on potentiostatic holds, which need a relatively short testing time. We suggest researchers carefully consider the voltage hold testing conditions set by the US DOE Silicon Consortium Project and the considerations reported by Schulze et al.⁶⁷ and Kalaga et al.⁶⁹ Briefly, there are three variables to consider for reliable qualitative evaluation of the voltage hold test. First, the cathode electrode must be designed with sufficient active lithium ions (for example, N/P ratio <1), with negligible potential variation, to prevent exhaustion of the active lithium ion inventory for reliable anode potential profiles. For this reason, the LiFePO₄ cathode is highly recommended as an ideal cathode for voltage hold testing because of its flat plateau. Second, qualitative voltage hold tests would be accurate only if sufficient hold time is applied to relax reversible lithiation and allow the hold test to be dominated by parasitic processes. Finally, the choice of voltage range is important because each of the tested Si anodes could have a different working potential.

Conclusions

Here we have demonstrated that seemingly decent performance parameters obtained for a Si anode with a certain specification does not guarantee its practical viability unless a thorough test is carried out that shows that the anode satisfies all the requirements discussed in our testing protocols. For example, some strategies can improve the cycle life of a Si anode, but they may not have the same benefit for the calendar life. It is therefore necessary to consider all practical requirements when developing Si anodes. We hope that this report will act as a guide for the development and evaluation of Si anodes with practical performance metrics. We also expect that it will have implications for the development of other next-generation battery anodes such as various alloy or conversion-type anodes and alkali metal (Li, Na and K) anodes, which share similar challenges to Si anodes.

Methods

Calculation of average anode voltage

The average voltages of NMC811, graphite and Si/SiO_x were set as 3.8 V (discharge), 0.13 V (delithiation) and 0.4 V (delithiation), respectively.

For blended anodes, we estimated the average voltage by a capacity-averaged linear combination of graphite and Si or SiO_x materials. The average anode voltage (V_{anode}) was calculated according to the following equation:

$$V_{\text{anode}} = \frac{C_{\text{Gra}}}{C_{\text{Si/Gra}}} \times 0.13 + \frac{C_{\text{Si}}}{C_{\text{Si/Gra}}} \times 0.4 \text{ (V)} \quad (1)$$

where C_{gra} , C_{Si} and $C_{\text{Si/Gra}}$ represent the specific capacity of the graphite electrode, the Si or SiO_x electrode and the Si-containing graphite electrode, respectively.

The average voltage of each full cell ($V_{\text{full cell}}$) was calculated as the difference between the average voltage of the cathode (V_{cathode}) and the average voltage of the anode:

$$V_{\text{full cell}} = V_{\text{cathode}} - V_{\text{anode}} \quad (2)$$

Calculation of energy density

The energy density and gravimetric energy of each cell were calculated according to the following equations:

$$\text{Energy density (Wh l}^{-1}\text{)} = \frac{V_{\text{average}} \times C_{\text{cell}}}{\text{Cell volume}} \quad (3)$$

$$\text{Gravimetric energy (Wh kg}^{-1}\text{)} = \frac{V_{\text{average}} \times C_{\text{cell}}}{\text{Cell weight}} \quad (4)$$

$$\begin{aligned} \text{Cell volume (l)} \\ = \text{Cell thickness at fully charged state} \times \text{Cell width} \times \text{Cell height} \end{aligned} \quad (5)$$

where V_{average} and C_{cell} represent the average voltage of a full cell and the total capacity of each cell, respectively. The cell weight is the total weight of the positive and negative electrodes, separator, casing and electrolyte in a cell.

Calculation of electrode porosity

The electrode porosity can be estimated from the following equation:

$$\text{Electrode porosity (\%)} = 1 - \frac{\text{Bulk density of electrode}}{\text{Theoretical density of electrode}} \quad (6)$$

The theoretical density of an electrode is calculated by considering the true density of each material in the electrode.

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Competing interests

The authors declare no competing interests.

Additional information

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